

Lab Exercise 17

HIGH LEVEL PROGRAMMING EXTENSIONS – FUNCTIONS

AIM

To implement **user-defined functions and recursive procedures** in MySQL using the `Vegetable` database.

PROCEDURE

1. Select the `VegetableDB` database.
2. Create a function to determine the **price level** of a vegetable based on its price.
3. Call the function using a vegetable price.
4. Set the recursion depth for recursive procedures.
5. Create a recursive procedure to calculate the factorial of a number.
6. Execute the procedure and display the result.

Problem 1: Function to Find Vegetable Price Level

Logic

- Price > 50 → **PLATINUM**
- Price between 30 and 50 → **GOLD**
- Price < 30 → **SILVER**

Create Function

```
USE VegetableDB;

DELIMITER //

CREATE FUNCTION GetVegetableLevel(price DECIMAL(10,2))
RETURNS VARCHAR(20)
DETERMINISTIC
BEGIN
    DECLARE vegetable_level VARCHAR(20);

    IF price > 50 THEN
        SET vegetable_level = 'PLATINUM';

    ELSEIF price >= 30 AND price <= 50 THEN
        SET vegetable_level = 'GOLD';

    ELSE
        SET vegetable_level = 'SILVER';
    END IF;
END IF;
```

```
        RETURN vegetable_level;
END //
```

DELIMITER ;

Execute Function

```
SELECT GetVegetableLevel(45.50) AS Vegetable_Level;
```

```
mysql> SELECT GetVegetableLevel(45.50) AS Vegetable_Level;
+-----+
| Vegetable_Level |
+-----+
| GOLD           |
+-----+
1 row in set (0.00 sec)
```

Another example:

```
SELECT GetVegetableLevel(25.00) AS Vegetable_Level;
```

```
mysql> SELECT GetVegetableLevel(25.00) AS Vegetable_Level;
+-----+
| Vegetable_Level |
+-----+
| SILVER          |
+-----+
1 row in set (0.00 sec)
```

Problem 2: Recursive Procedure to Calculate Factorial

Set Recursion Limit

```
SET @@SESSION.max_sp_recursion_depth = 255;
```

Create Recursive Procedure

```
DELIMITER //
```

```
CREATE PROCEDURE Factorial(
    IN n INT,
    OUT result BIGINT
)
BEGIN
    IF n <= 1 THEN
        SET result = 1;
    ELSE
```

```
        CALL Factorial(n - 1, @temp);
        SET result = n * @temp;
    END IF;
END //
```

DELIMITER ;

Execute

For factorial of 5:

```
CALL Factorial(5, @result);
SELECT @result AS Factorial;
```

```
mysql> SELECT @result AS Factorial;
+-----+
| Factorial |
+-----+
|      120 |
+-----+
1 row in set (0.00 sec)
```